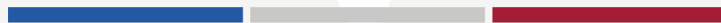




MIT SLOAN SCHOOL OF MANAGEMENT
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MODULE 3 UNIT 4
Podcast 1 Transcript

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Fifteen days in

PAUL MCDONAGH-SMITH: The third Friday was different from the first two. In the week between Episodes 2 and 3, the supplier qualification agent had gone live. Not at scale, not in the production sense, but live in the sandbox. Olu had defined: low value suppliers, low risk jurisdictions, human sign-off at every decision of consequence. Fifteen working days roughly a hundred and eighty supplier qualifications run through the agents. A small sample, but a real one. The team came into the third meeting with data. They also came in, as I would shortly learn, with a near miss.

The CEO began by asking Rebecca to open. Rebecca had a one-page summary in front of her. She said she would walk the team through what the pilot had taught them: the good, the useful, and, she said carefully, the alarming.

The good was genuinely good. The agent's cycle time on the qualifications it handled was on average 4 days compared with a baseline of 36. The agents' decisions on the cases that had also been reviewed by a human aligned with the human decision in roughly 92% of instances. Of the disagreements, most were ambiguity cases where either answer was defensible. The procurement specialist whom Rebecca and Priya had assigned to manage the agents, a senior analyst called Hannah (six years at Boulton Industries), had taken to the role in a way neither of them had expected. She had started proposing improvements after the first 10 days.

The useful was more interesting than the good. Rebecca said the AgentOps dashboard had surfaced patterns the team had not anticipated. The agent was unusually confident on certain supplier categories and unusually uncertain on others. The uncertainty clustered around suppliers whose documentation was in formats the agent hadn't seen much of during its test runs. Hannah had spotted this and flagged it, and the team had adjusted the threshold map in response. "The dashboard is not a report," Rebecca said. "It's a conversation we are now having continuously with our own system."

Then Rebecca came to the alarming. On day 11 of the pilot, the agent had been run over a supplier application from a jurisdiction the team had classified as low risk. The application appeared clean. The financials checked out. The certifications were valid. The agent's recommendation was to qualify the supplier for onboarding. But the sanctions flag layer, the hard coded rule Olu had insisted lived outside the model's reasoning, had halted the process. The supplier itself was not sanctioned. Their parent company, a level of ownership up, was under investigation by a regulator in a different jurisdiction and that investigation had been added to a watch list two days earlier. The agent's perception layer by itself had not connected the two. The hardcoded check had.

Had that check not been there, the agent would have qualified the supplier. Boulton would, in due course, have placed business with an entity whose parent was facing active regulatory action. The reputational and compliance consequences wouldn't have been trivial. "Nobody was harmed," Rebecca said. "The safeguard worked. But I want to put the question on the table because it's the scaling question. What if the threshold had been one

notch to the right? What if this agent had been one of 20, each bounded slightly differently, each running at slightly different levels of autonomy? What is the mechanism that catches this in every instance, across every deployment, every day, because that is the question the board is going to ask?”

The room went quiet. The CEO let the silence do its work. And then she said, “That is exactly the right question. Let’s spend the day on it.”

The paradox of scaling

MCDONAGH-SMITH: Vikram began the substantive discussion with a reframing that I’ve since come to think is an important conceptual move of this three-meeting series. “There is a paradox in how most organizations scale agentic systems,” he said. “It goes like this: You build a pilot. The pilot works. Stakeholders get excited. The pressure builds to scale, to do more of what is working, faster. The temptation is to treat scaling as an extension of the pilot. Take what works and do it in more places.” He paused.

“That is not what scaling is, at least not for this class of technology. Scaling an agentic system is not enlarging a pilot. It’s transitioning from a controlled environment where a small team understands every decision the agent makes to an operational context where the agent acts across functions, across data sources, across situations that no one has explicitly tested. The failure modes change, the governance requirements intensify. The measurement problem deepens and the organizational dynamics shift, because scaling means the technology now touches people who did not volunteer for the pilots.” Olu underlined a sentence in her notebook. She showed it to me later. The sentence was “Scaling is a transition, not an extension.”

Vikram continued, “The other thing I want to warn us about is what a researcher I respect calls the ‘seduction of the demo.’ These models make it extraordinarily easy to build impressive demos quickly. The demo works in the controlled setting. Stakeholders are enamored with it. Resources get allocated. And then deployment begins and you discover the corner cases. The edges where the agent hallucinates, selects the wrong tool, misreads the intent, propagates an error. The distance between the demo and what’s ready for production at scale is usually larger than anyone expects. And we’ll be tempted to close that distance with optimism. We should close it with design.”

And then he said something that sent the meeting in the direction it needed to go. He said, “I don’t think our scaling problem is primarily a technology problem. The models will keep improving. Our plumbing will keep improving. The bottleneck at scale will be governance.” Olu said, “Good, because I’ve been thinking about governance all week and I want to introduce a concept that I think the team needs to hold in its head before we go any further. This concept is autonomy, but autonomy understood properly.

She walked to the whiteboard. She drew a horizontal line. At the left hand end, she wrote “Fully constrained.” At the right hand end, she wrote “Fully autonomous.” In between, she drew no markings at all. “Most people, when they hear the word autonomy, think of it as a switch,” Olu said. On or off. Either the agent is autonomous or a human is in control. I want to propose that this is the wrong mental model. Autonomy isn’t a switch. It’s a dial. And the critical question in every deployment, every single one, is where on the dial you set it, for this task, in this context, with these stakes, against these safeguards.”

She gave two examples: A fully constrained agent could only select from a pre verified set of known correct responses. It could not hallucinate because it wasn't generating anything novel. It was safe, so to speak. But it was also limited. A fully autonomous agent could generate novel outputs, take actions in external systems, and execute multistep plans without human intervention. Extraordinarily powerful. Every output carrying uncertainty.

"Most deployments," Olu said, "should sit somewhere in between. And the portfolio question, the one the board is asking us, is not where on the dial does Boulton Industries sit. It's where on the dial does each deployment sit and how do we govern the fact that we will have 20 different positions on that dial arguably at the same time." That reframing unlocked the meeting.

Four layers, five people

MCDONAGH-SMITH: What the team did next was, I think, perhaps the most elegant collective act of design I'd seen over the three weeks. Olu said she wanted to name four layers of safeguard that any agentic system at scale should have in place. She said she would introduce each one briefly, but that she wanted a different person in the room to speak to each layer, because the safeguard architecture, properly understood, was distributed across the organization not owned by any one function.

The first layer, she said, was hardcoded constraints. These were nonnegotiable boundaries built into the system architecture itself. Rules that executed outside the model that the model could not talk its way past. The sanctions watchlist check that had caught the parent company issue on day 11 was a hardcoded constraint. Olu spoke to this layer herself because it was legal in origin: "The hardcoded constraints are how we translate our nonnegotiables into code. They aren't prompts. They are not guidance to the model. They are gates that execute independently. If we are going to run 20 of these, we need a registry of hardcoded constraints that applies across every deployment and a discipline for updating it when the world changes.

The second layer, she said, was output verification. Before an agent's output reached a user or triggered a downstream action, it should be checked against formal rules, consistency criteria or a second system acting as a reviewer. She turned to Maria. "This one sits with operations. Maria, you do this in manufacturing every day. What does output verification look like in a factory?"

Maria picked it up cleanly, "In manufacturing, we call it two-person rule. Nothing consequential happens on the word of one person or one system. A critical decision requires a second check, sometimes by a different person, sometimes by an independent instrument. The point isn't redundancy for its own sake. The point is that you detect errors at the boundary of the act, not after it. For agents at scale, this means building a verification layer. Probably a second agent or a deterministic checker or both between the agent's decision and its action. The verification is not optional in high-stakes contexts. It is the architecture."

The third layer, Olu said, was reversibility by design. Wherever possible, an agent's actions should be designed to be undoable. If an agent sent a communication, could it be

recalled? If it initiated a transaction, was there a rollback window? She turned to Rebecca. "This one belongs to finance, I think."

And Rebecca took it up, "In financial controls, we have a concept called blast radius. It's the amount of damage a single decision can do before it's caught. Good, sound financial architecture shrinks the blast radius. Not every decision can be made reversible. Some are, by their nature, committed the moment they are made. But a surprising number can be made reversible if you design for it. Time-delayed execution, soft commits that become hard after a review window, segregated accounts that limit exposure. The discipline I would want to apply to agents is never let an agent make an irreversible decision faster than our ability to detect that it was wrong. That is a principle I can operationalize."

And then the fourth and final layer, Olu said, was escalation architecture. Every agent should have a clear tested escalation path. A mechanism for recognizing that it had reached the boundary of its confidence or its authority and routing the decision to a human. The design challenge here, she said, was calibration. Escalate too often and you have rebuilt a manual process with extra steps. Escalate too rarely and you have an autonomous system operating beyond its competence. She turned to Priya.

Priya said, "This is the layer I care most about, because it's the layer where humans actually meet agents. And it's the layer where most organizations get the design wrong. The common failure mode is to treat escalation as a fallback, a safety valve for the one-in-a-thousand cases. But it's not that. It's the primary interface between the agentic system and the people who are accountable for its decisions. Which means the escalation path has to be staffed. It has to be trained. It has to be funded, to be given real authority, and crucially – this is the Hannah point from the pilot – the person on the receiving end of the escalation has to have enough context to actually make the decision. An escalation without context isn't an escalation. It's a handoff of risk."

And Vikram had been listening. He said, "The four layers work as a defense in-depth. No single layer is sufficient. Every serious agentic deployment requires all four and the architectural discipline is that each layer is independent. The safety of the whole system doesn't rely on any single layer working perfectly because we know no single layer ever will."

And so I watched the team get still around that observation. What had emerged on the whiteboard wasn't a governance policy. It was an architecture. Each function in the room now owned a piece of it. The safeguard layers were distributed across the leadership team in the same way they would need to be distributed across the organization. The CEO said this is the piece the board will want to see.

The comparison that matters, and the horizon beyond it

MCDONAGH-SMITH: Rebecca brought the conversation back to measurement. But she brought it back differently from how she had framed it in Episode 2. "In the second meeting," she said, "we talked about measuring the agent against the existing process. I want to sharpen that now because the frame I've seen executives use most often, including, I confess, the frame I started with, is to ask whether the agent is perfect. And it is not. No agent will be. Some percentage of the decisions this system makes will be wrong. We need to be honest about that."

She paused. “But the comparison is not the agent versus perfection. The comparison is the agent versus the current process. And the current process, our own, I mean, over the last 10 years of supplier qualification isn’t perfect either. Humans make mistakes. Inconsistently, unpredictably, often without an audit trail. The question isn’t whether the agent makes errors. The question is whether this agent, with these safeguards, produces better outcomes than the process it replaces, on the dimensions that matter: accuracy, consistency, traceability, auditability, speed, capacity released.”

She looked around the table. “On every one of those dimensions, 15 days in, the agent is at or above the current process. And because it’s consistent, because everything it does is logged, because every decision can be traced, it’s actually outperforming the current process on the dimensions the regulator cares about most. That’s the comparison I would put in front of the board.”

The CEO said, “That’s the ROI narrative. Write that down.” Rebecca had already written it down. Then Vikram did something I’d been quietly hoping he would do and which the team I think had perhaps half forgotten was available to them. He went back to the slide he had shown in the first meeting. The continuum with the blank column at the far right. He put it up on the screen again. And he said, “I want to tell you what I think is going to happen over the next two to three years. Not because it changes what we do today, because it should change what we invest in.”

He described a trajectory he and colleagues had been tracking in the research community: Historically, if you wanted a computer system to do something, you wrote code: precise, syntactically exact instructions in a formal programming language. Each generation of programming language had raised the level of abstraction, from machine code to assembly, to high level languages, to declarative languages like SQL, where you specified what you wanted and the system figured out how to deliver it. “The next step,” Vikram said, “points to a further leap. Systems where the specification is not written in a programming language at all. However, in a structured natural language. You describe in detailed rigorous prose what the system should do, how it should behave, what its constraints are, and agents generate the code. They deploy it, they maintain it, they regenerate it as the models improve.”

He paused. “If that horizon is real, and I think it is, the implications for Boulton Industries are significant. In that world, the source code is no longer the intellectual asset. The specification is. Our business logic, our constraints, our processes, our rules, written clearly enough that an agent could build the system that implements them. I’m not asking us to act on this today. I’m asking us to notice that the assets we should be building and the assets we should be protecting might not be the ones we’ve assumed.

Maria then asked the question I was waiting for someone to ask, “What does that mean we should do now?” Vikram replied, “It means we should treat our process documentation, our data quality, and our articulated business logic as competitive assets, not administrative overhead. It means that structured knowledge, clean data, and well-defined processes will appreciate in value. Rigid custom code and tightly coupled vendor dependencies will depreciate. I would add a line to the board recommendation that says so.”

The CEO thought about that for a long moment. And then she said, “Add the line, but frame it as an horizon, not a commitment. We are not betting the company on a two, three, five

year forecast. We are acknowledging that our investments in data and process documentation are also now investments in our agentic future.” That, I noted, was the moment the blank column on Vikram’s slide stopped being a placeholder and became part of Boulton’s strategic arguments.

The opening sentence

MCDONAGH-SMITH: The last hour of the third meeting was spent drafting the board recommendation. I’m not going to walk you through the document itself because its detail is specific to Boulton Industries and not really transferable. But I do want to tell you how it opened because the opening sentence, the one the CEO composed at the whiteboard with input from all five of her colleagues, is the sentence I would want to leave you with.

Before she got to it, Priya said one more thing that I think the recommendation needed. “I’ve been reading Darron Acemoglu,” Priya said, “The Nobel laureate. The economist at MIT. And he argues across a thousand years of technological history that technology does not automatically distribute its benefits. Sometimes it has, sometimes it hasn’t. The difference between the periods of broadly shared prosperity and the periods of concentrated extraction, he argues, is institutions. The rules, the norms, the governance structures, the countervailing power that channel technology toward human complementarity rather than human displacing applications. Agentic AI is a test of whether we can build those institutions inside our own organization.”

She looked at the CEO. “I’d like the Board recommendation to acknowledge that, not as a political statement, but as a strategic one, because governance is not the constraint on what we can do. It’s what makes what we can do durable.” The CEO agreed. The phrase governance as a differentiator, not a constraint went into the document. It became the organizing principle of the closing section.

And then Olu raised the last of the architectural points. And it was one we hadn’t yet discussed. “Agents,” she said, “potentially have identities. They have permissions. They access systems. They take actions. Each of those things has historically been tied to a human user. A human user who could be held accountable, whose access could be traced, whose permissions could be revoked. Agents break that assumption. We are going to need to manage nonhuman identities the way we currently manage human ones. Each agent will need its own identity, its own least-privileged access, its own audit trail. This is a discipline that most organizations haven’t built yet. We should build it from the start.”

That went into the document too. And then, just before eleven o’clock, the CEO wrote the opening sentence of the board recommendation on the whiteboard. It read, “Boulton Industries should deploy agentic AI, deliberately and at a measured pace, in a manner that treats governance, safeguard architecture, and organizational readiness as the central design variables rather than downstream constraints, because it’s in the quality of those variables that the durability and the differentiation of our agentic capability will be found.”

Vikram read it aloud. Rebecca adjusted two words. Olu added a subordinate clause about the autonomy dial. Priya asked for the insertion of the people who manage these systems. Maria said the sentence was now slightly too long and could lose the phrase “at measured pace.” The CEO accepted every edit except Maria’s. “The pace language stays,” she said. “It’s the honest description of what we are proposing.” The meeting closed. The team dispersed to finish drafting. The board would have the documents the following Monday.

Key takeaways

MCDONAGH-SMITH: So let me offer a closing reflection as the person who had sat through all three of these meetings. What I watched over those three Fridays was an executive team take a difficult, contested, high-consequence question and work through it with unusual honesty. They didn't pretend to have answers they didn't have. They didn't rush to conclusions to meet a deadline. They used the language they needed and revised it when it didn't serve them. They acknowledged, all six of them at different moments, that the organization they led was not yet the organization it would need to become. And they designed their way towards it, not by adopting a vendor's blueprint, but by naming their own.

And if I had to compress what the third meeting surfaced into a short set of propositions, the kind a listener could carry back into their own organization, it would be these:

- Scaling is a transition, not an extension – a move from controlled environment to operational context, with qualitatively new failure modes.
- Autonomy is a dial, not a switch and the governance question is where every deployment sits on that dial and how the portfolio is managed as a whole.
- The four safeguard layers: hardcoded constraints, output verification, reversibility by design, and escalation architecture are a defense in-depth, each independent of the others, distributed across the functions that can actually own them.
- The comparison that matters is never agent versus perfection. It's agent with safeguards versus the current process.
- On the dimensions that matter: The horizon ahead rewards organizations that treat structured knowledge, clean data, and articulated business logic as strategic assets.
- Nonhuman identities, agents with access, with permissions, and actions require the same levels of discipline of identity management we've built over decades for human users.
- And governance, done well, is not a constraint on agentic AI. It's what makes agentic AI durable.

Those propositions are not the recommendation Boulton made. They are what the recommendation was built upon.

And I want to leave you, as your program does, with an invitation: Somewhere in your organization there is a question the board is either asking or about to ask that looks like the one Boulton Industries was given three Fridays before this podcast ends. How should we respond to agentic AI?

The answer to that question is not a product. It's not a platform. It's not a vendor selection. The answer to that question is a recommendation. A document, perhaps a short one, written by a group of people who have worked honestly through the vocabulary, the

architecture, and the governance. A recommendation that makes the case for deploying agentic AI in a specific way, with a specific scope, against a specific set of safeguards, for a specific kind of organizational value. And the program you've just completed is designed to help you write that recommendation for your organization in your voice with your board as the reader. The capstone assignment asks you to do exactly that.

The Boulton industry team had three Fridays. You, I suspect, will have more time than that, or perhaps less. What you will not have is the option to wait until the landscape holds still. Because it shan't. So write your opening sentence. Revise it. Let your colleagues edit it. Put the pace language back in when someone tries to take it out, and then send the document to the board. That's the work and it is now your turn to undertake it.

Thank you.